Infoviz term: channels

When you represent data using graphics, you need to consider the best visual methods to represent the types of data you're working with. Each graphical object, as well as the whole display, can be broken down into component channels that relay information visually. These channels, such as height, width, area, color, position, and shape, are particularly well suited to represent different classes of information. For instance, if you represent magnitude by changing the size of a circle, and if you create a direct correspondence between radius and magnitude, then your readers will be confused, because we tend to recognize the area of a circle rather than its radius. Channels also exist at multiple levels, and some techniques use hue, saturation, and value to represent three different pieces of information, rather than just using color more generically.

The important thing here is to avoid using too many channels, and instead focus on using the channels most suitable to your data. If you aren't varying shape, for instance, if you're using a bar chart where all the shapes are rectangles, then you can use color for category and value (lightness) to represent magnitude.

Going back to the tweets.json data, it may seem like there's not much data available to put on a chart, but depending on what factors we want to measure and display, we can take a couple different approaches. Let's imagine we want to measure the impact factor of tweets, treating tweets that are favorited or retweeted as more important than tweets that aren't. This time, instead of a bar chart, we'll create a scatterplot, and instead of using array position to place it along the x-axis, let's use time, because there's good evidence that tweets made at certain times are more likely to be favorited or retweeted. We'll place each tweet along the y-axis using a scale based on the maximum impact factor of our set of tweets. From this point on, we'll focus on the `dataViz()` function as in the following listing, because you should be familiar now with getting your data in and sending it to such a function.

Listing 2.5 Creating a scatterplot

```
function dataViz(incomingData) {

incomingData.forEach(function (el) {
  el.impact = el.favorites.length + el.retweets.length;
  el.tweetTime = new Date(el.timestamp);
})

var maxImpact = d3.max(incomingData, function(el) {return el.impact;});
var startEnd = d3.extent(incomingData, function(el) {
  return el.tweetTime;
});
var timeRamp = d3.time.scale().domain(startEnd).range([20,480]);
var yScale = d3.scale.linear().domain([0,maxImpact]).range([0,460]);
var radiusScale = d3.scale.linear()
                    .domain([0,maxImpact]).range([1,20]);
```

Creates an impact score by totaling the number of favorites and retweets

Transforms the ISO 8906–compliant string into a date datatype

Returns the earliest and latest times for a scale

startEnd is an array.